Methodological Description and Column Reference Guide

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1. Introduction

This document provides a concise overview of the curated metadatabase catalog developed within the GEOS-EUDR project, Thünen Institute for Forestry. The catalog compiles published and publicly available geospatial datasets related to forests, commodities, and other land-use themes across global, regional, and selected national scales.

It supports spatially explicit analyses under the European Regulation on Deforestation-free Products (https://environment.ec.europa.eu/topics/forests/deforestation/regulation-deforestation-free-products_en), with a focus on improving access to relevant datasets for monitoring deforestation-free supply chains and land-use dynamics.

Within this framework, the metadatabase organizes and documents datasets in a structured manner to facilitate transparent discovery, comparison, and reuse. It is designed to support both research and policy applications related to forest monitoring, land-use analysis, and sustainability assessments across multiple spatial scales.

2. Methodological Description

The dataset search and compilation for this catalog has been ongoing since May 2024, with periodic updates to Version 1 continuing through May 2026. Further continuous updates may be incorporated in subsequent versions, as new relevant datasets become available.

Dataset Search Strategy

The dataset search was carried out through a comprehensive review of publicly available global geospatial datasets providing information on forest and tree cover derived from Earth Observation (EO) and remote sensing data. The search covered major geospatial data platforms and repositories, including Google Earth Engine, USGS Earth Explorer, NASA EarthData, ESA Copernicus data services, Global Forest Watch, FAO GeoNetwork, the US Open Data Portal, UNEP Environmental Data Explorer, ArcGIS Living Atlas, UNdata, and World Bank Open Data.

To complement this database search, additional datasets were identified through cross-referencing with existing synthesis reports, including the Forest Resources Assessment (2020), as well as documentation related to digital public infrastructure supporting deforestation-related trade regulations. Furthermore, a structured literature review was conducted using scientific search engines such as Google Scholar, Scopus, and Web of Science to identify peer-reviewed publications and globally relevant datasets related to forest cover, tree cover, and land use/land cover mapping.

This multi-source approach ensured a broad and systematic coverage of available datasets relevant for global forest and land cover monitoring applications.

Dataset Inclusion Criteria

Datasets were included in the catalog only if they contain information relevant to at least one of the following thematic elements:

- Forest cover
- Tree cover
- Land use / land cover
- Forest degradation or related proxies
- Tree plantations or planted forests
- Deforestation or tree cover loss
- EUDR-relevant commodity production systems (cattle, wood, cocoa, soy, palm oil, coffee, and rubber)
- General crop extent or agricultural land use

Temporal and Spatial Criteria

To ensure comparability and relevance across datasets, the following constraints were applied:

- Temporal coverage: datasets must fall within approximately ±5 years around the EUDR reference year 2020
- Spatial resolution: datasets must have a spatial resolution of 30 m or finer

Datasets that do not satisfy these criteria were excluded from the catalog.

Additional Notes on Forest Definition

For datasets related to forest and tree cover, the applied definition of the "forest" class is explicitly documented within the catalog. This includes, where available, information on thresholds such as canopy cover, minimum mapping unit, and vegetation height.

In 2025, a study (*Freitas Beyer et al., 2025*, https://www.mdpi.com/2072-4292/17/17/3012) published that describes the search for and selection of global datasets on forest and tree cover for the purpose of the EUDR suitability assessment, as well as a comparison of the relevant datasets.

3. Column Reference Guide

Item-ID	Unique identifier assigned to each dataset entry.
Dataset Name, Abbreviation and version(s)	Full name of the dataset and its commonly used abbreviation (if applicable), along with available versions.
Data Provider / Responsible Organization	Institution or organization responsible for dataset development or maintenance.
Primary Publication / Citation	Main scientific publication or reference associated with the dataset.
Link to Publication/Technical Report	URL to supporting documentation, reports, or peer-reviewed publications.
Repository Link (GEE)	Link to dataset availability in Google Earth Engine, if applicable.
Repository Link (Other)	Alternative repository or platform hosting the dataset, if available.
Mapped Classes / Class Definitions	Description of land cover or thematic classes used in the dataset.
Forest/ Tree Cover Definition (Thresholds: Height; Minimum area; Canopy cover) *Only for forest/ tree cover-related datasets	Specific definition of forest or tree cover classes, including quantitative thresholds where available.
Extent (Global / Regional / National)	Geographical extent of the dataset.
Spatial Resolution (m)	Spatial resolution of the dataset in meters.
Start Year	First year of data availability.
End Year	Last year of data availability.
Temporal Resolution	Frequency of observations (e.g., annual, seasonal, monthly).
Sensor / Primary Data Source	Satellite sensor(s) or primary data source used.
Input Metric/ Indices/ Features	Variables or features used as inputs (e.g., spectral indices, biophysical variables).
Classification Method(s) (Specific)	Summarized description of the classification algorithm(s) applied.
Classification (Category)	General classification approach used, such as supervised machine learning, deep learning, rule-based/ threshold-based, expert-driven, unsupervised, hybrid methods, or visual interpretation.

Number of Training Samples	Number of samples used for model training, if applicable.
Validation Approach	Method used for dataset validation, if applicable.
Reference Dataset for Validation (Data Name and Period)	External dataset(s) used for validation and their temporal coverage.
Number of Testing and/ or Validation Samples	Number of samples used for testing or validation.
Overall Thematic Accuracy (OA)	Overall accuracy of the dataset classification.
Producer's and User's Accuracy (target class)	Accuracy metrics for specific classes, if available.
Other Accuracy Metric(s)	Additional performance metrics reported by dataset providers, if available.
Additional Accuracy Information	Additional Accuracy Information, if available.
Reported Limitations	Known limitations, uncertainties, or constraints associated with the dataset, if available.

Disclaimer: Not all datasets include information for every column listed above.